

Seminar Speaker Script

Sparse Recovery and Compressed Sensing: A Model-Based Deep Learning Approach

Model-Based Deep Learning (361.2.2320) — Recorded seminar, ~15 minutes

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Slide 1 — Title

0:00–0:30

Hello everyone. We are Etay Baron and Etai Wigman, and this is our final project for the Model-Based Deep Learning course. Our topic is sparse recovery and compressed sensing, based on the NeurIPS 2021 paper “Hyperparameter Tuning is All You Need for LISTA.” The core question that guides this talk is simple: how far can model-based structure take us — and when do we need data-driven flexibility?

Slide 2 — Seminar Roadmap

0:30–1:10

Here is the plan for the next fifteen minutes. We start with the background: the sparse recovery problem, the LASSO formulation, and the classical ISTA and FISTA algorithms. We then move to deep unrolling — LISTA, ALISTA, and HyperLISTA. The central part is our own study: we ask which components of ISTA are actually worth learning, and we compare the training methods for unfolded optimizers that were discussed in class. Finally, we stress-test everything on real images — MNIST and Fashion-MNIST — under compressed sensing. Our synthetic setup uses 250 measurements, signal dimension 500, and sparsity 50.

Slide 3 — Motivation & Problem Formulation

1:10–2:10

Let us define the problem. We want to recover a sparse vector x^* of dimension n , but we only observe m linear measurements, with m smaller than n , possibly with noise. This inverse problem is underdetermined — infinitely many signals explain the same measurements. The sparsity assumption is what makes it solvable, and this situation appears everywhere: MRI acceleration, radar, hyperspectral imaging. The standard formulation is the LASSO: a trade-off between fidelity to the measurements and an ℓ_1 norm that promotes sparsity. Our metrics are NMSE in decibels for synthetic signals, plus PSNR and SSIM for images.

Slide 4 — Classical Baselines: ISTA and FISTA

2:10–3:10

ISTA is proximal gradient descent on the LASSO objective. Each iteration takes a gradient step on the data-fidelity term, and then applies soft-thresholding, which shrinks small entries to zero. FISTA adds Nesterov momentum and improves the convergence rate from order $1/k$ to $1/k^2$. But here is the key point: after only sixteen iterations, ISTA reaches just minus

5.3 dB, and FISTA minus 11. Reaching minus 20 dB takes hundreds of iterations. So the question is: how do we make sixteen steps really count?

Slide 5 — Deep Unrolling: From Iterations to Layers

3:10–4:20

The idea, due to Gregor and LeCun in 2010, is to unfold the K iterations of ISTA into a network with K layers, and to turn the algorithmic constants into learnable parameters. On the left you see classical ISTA, with a fixed step size and a fixed threshold. On the right is LISTA: two weight matrices and a separate threshold per layer, all trained end-to-end with backpropagation. This is the model-based deep learning principle: keep the structure — a gradient step followed by soft-thresholding — and learn only what improves reconstruction at a finite depth. In our setting this amounts to about six million parameters.

Slide 6 — The MBDL Ladder: LISTA $\rightarrow$ ALISTA $\rightarrow$ HyperLISTA

4:20–5:50

Now the MBDL ladder — more structure, fewer parameters. LISTA learns everything: six million parameters, minus 20 dB. ALISTA shows that the weight matrix does not need to be learned at all: it can be computed analytically from A , leaving only per-layer step sizes and thresholds — 32 parameters, and minus 30 dB. HyperLISTA takes this to the extreme: only three scalars. The threshold, the momentum, and the support selection are computed adaptively from the residual at every layer, using the formulas on the slide. And notice — there is no backpropagation at all: the three scalars are found by a simple grid search. The result: minus 61 dB, the best of all. More structure, fewer parameters, better performance — this is the MBDL principle in action.

Slide 7 — Our Design Space Study: What Should We Learn?

5:50–6:50

This leads to our research question: between six million parameters and three, what is the minimal set that is actually worth learning? The generalized ISTA update exposes two free scalar sequences: the step size γ and the threshold θ . We designed three models. ThresholdISTA learns only the thresholds. StepISTA learns only the step sizes, with the threshold tied to them. And StepThresholdISTA learns both, independently — 32 scalars in total. All of them are initialized to the exact ISTA values, so before training, every model is identical to ISTA.

Slide 8 — Our Results: Scalar Models

6:50–8:00

The results are quite striking. Learning thresholds alone barely helps: minus 7 dB. But learning the step-size schedule alone — just sixteen scalars — jumps to minus 20 dB and matches the full LISTA with six million parameters. Decoupling the threshold from the step adds another 3.6 dB, essentially for free. The conclusion: the step-size schedule is the critical component to learn at finite depth, and learning full matrices is largely redundant when the

sensing matrix is fixed.

Slide 9 — Training Methods for Unfolded Optimizers

8:00–9:10

Following the training strategies discussed in class, we compared them on the exact same architecture. L1 is standard end-to-end training, with a loss on the final layer only. L2 is deep supervision: a weighted loss on every intermediate layer. L3 is greedy layer-wise training, where each layer is trained while the previous ones are frozen. We also tested weight tying — the same layer repeated K times, like a recurrent network. The interesting finding: tied LISTA, with sixteen times fewer parameters, comes very close to the untied version. Weight tying acts as a built-in regularizer.

Slide 10 — Synthetic Sparse Recovery Results

9:10–10:10

Here is the full picture on the synthetic problem: NMSE as a function of depth. The training method matters — L2 is the best, and L3 gets stuck around layer eight, because later layers have no incentive to improve on what earlier layers already solved. But the big hierarchy remains: ALISTA beats every LISTA variant with 32 parameters, and HyperLISTA beats everyone with three. When the model assumptions hold, structure beats capacity.

Slide 11 — Design-Space Map

10:10–11:00

This map summarizes the synthetic part. The x-axis is the number of parameters on a logarithmic scale; the y-axis is the reconstruction error; the bottom-left corner is the ideal. Our StepThresholdISTA, with 32 scalars, sits in an excellent spot and beats LISTA with 187,000 times fewer parameters. HyperLISTA breaks the scale entirely. The message: in problems with known structure, choosing what to learn matters more than raw capacity.

Slide 12 — Real Image CS: FashionMNIST Pixel Domain

11:00–12:10

Now the real test — actual images. We moved to pixel-domain compressed sensing on Fashion-MNIST: the image is flattened into a 784-dimensional vector, and we measure only a quarter of that dimension. No transform is needed, because these images have a black background and are naturally sparse in pixels. And here the ranking flips completely. All the structured methods collapse — ALISTA and HyperLISTA do even worse than plain ISTA — while LISTA with deep supervision reaches an SSIM of 0.82. The reason: the assumption of i.i.d. Gaussian sparsity is simply wrong for bounded, spatially-correlated pixels. A wrong prior is worse than no prior.

Slide 13 — Robustness Across Datasets and Measurement Ratios

12:10–13:10

We tested this systematically over two datasets and two measurement ratios; the heatmap

shows SSIM. On MNIST with half the measurements, every method does fine, because MNIST is genuinely sparse — about 78 percent of its pixels are zero. As the task gets harder — fewer measurements, or more textured data — the structured methods degrade sharply, while LISTA-L2 stays above 0.82 in every scenario. The sparse prior breaks down gradually as the data moves away from the assumptions.

Slide 14 — Visual Stress Tests

13:10–14:00

The visual comparison makes the numbers concrete. MNIST is on the left, Fashion-MNIST on the right, both with a quarter of the measurements. The LISTA-L2 row is almost identical to the originals in both cases. ALISTA and HyperLISTA literally erase real signal — a threshold calibrated for sparse Gaussian residuals deletes the image itself. This is what a wrong prior looks like.

Slide 15 — Conclusions

14:00–14:50

To summarize, four lessons. One: when the assumptions hold, structure wins — three scalars beat six million parameters by 41 dB. Two: the step-size schedule is the most valuable thing to learn. Three: weight tying is implicit regularization. Four: a wrong prior actively hurts — on real images, the flexible LISTA wins. As future work, we would like to test these models in the DCT or wavelet domain, where images are much closer to the Gaussian-sparse assumption.

Slide 16 — Code & References

14:50–15:00

All of our code — the four notebooks and the source modules — is available in a public GitHub repository, and these are the main references we built on. Thank you very much for listening.